



Tech Info Library

Applesoft: Internals--Floating Point Math Package (2 of 3)

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Security: Everyone

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FLOATING POINT REGISTER MOVE ROUTINES

MOVFM EAF9
Moves the number in memory Y,A points to into FAC. On exit A and Z reflect FACEXP.

MOV2F EB1E
Pack FAC and temporarily moves it into register 2. On exit A and Z reflect FACEXP.

MOV1F EB21
Pack FAC and moves it temporarily into register 1. On exit A and Z reflect FACEXP.

MOVML EB23
Pack FAC and move it into zero page area starting at X. On exit A and Z reflect FACEXP.

MOVMF EB2B
Pack FAC and move it into memory pointed to by Y,X. On exit A and Z reflect FACEXP.

MOVFA EB53
Move ARG into FAC. On exit A and Z reflect FACEXP.

MOVAF EB63
Move FAC into ARG. On exit A and Z reflect FACEXP.

CONUPK E9E3
Load ARG from memory pointed to by Y,A. On exit A and Z reflect FACEXP.

SUMMARY OF MOVES:

FAC	=> (Y,A)	EB2B	MOVMF
FAC	=> (0,X)	EB23	MOVML

FAC	=>	TEMP1	EB21	MOV1F
FAC	=>	TEMP2	EB1E	MOV2F
FAC	=>	ARG	EB63	MOVAF
ARG	=>	FAC	EB53	MOVFA
(Y,A)	=>	FAC	EA69	MOVFM
(Y,A)	=>	ARG	E9E3	CONUPK

FLOATING POINT OPERATORS

The following routines require that A and Z reflect FACEXP. Even though FAC move routines set up A and Z, a LDA \$9D will insure their proper values.

FMULT E97F
Multiply the FAC by the number in memory pointed to by Y,A.

FMULTT E982
Multiply FAC and ARG. On entry A and Z must reflect FACEXP.

FDIV EA66
Divide the number in memory pointed to by Y,A by FAC.

FDIVT EA69
Divide ARG by FAC. On entry A and Z must reflect FACEXP.

FADD E7BE
Add the number Y,A points to in memory to FAC.

FADDT E7C1
Add FAC and ARG. On entry A and Z must reflect FACEXP.

FSUB E7A7
Subtract FAC from the number Y,A points to in memory.

FSUBT E7AA
Subtract FAC from ARG. On entry A and Z must reflect FACEXP.

TRANSCENDENTAL OPERATORS

FPWRT EE97
Raise ARG to the FAC power. On entry A and Z must reflect the value of FACEXP.

SQR EE8D
Takes the square root of FAC

LOG E941
Log base e of FAC

EXP EF09
Raise e to the FAC power

COS EFEA
Returns the cosine of the FAC in radians.

SIN EFF1
Returns the sine of the FAC in radians.

TAN F03A
Returns the tangent of the FAC in radians.

ATN F09E
Returns the inverse-tangent of the FAC in radians.

OTHER OPERATORS

SGN EB90
Modifies the value of the FAC depending on its value on entry. On exit:

FAC = 1 if FAC was greater than 0
FAC = 0 if FAC was equal to 0
FAC = -1 if FAC was less than 0

ABS EBAF
Absolute value of FAC. If FAC is less than 0 then FAC = -FAC.

INT EC23
Greatest integer value of FAC. Uses QINT and floats the result.

RND EFAE
Form a pseudo-random number in FAC and update RND at 00C9.

SIGN EB82
Set A according to the value of FAC. On exit:

A = 1 if FAC is positive
A = 0 if FAC = 0
A = FF if FAC is negative

FCOMP EBB2
Compare FAC and a packed number in memory pointed to by Y,A. On exit:

A = 1 if (Y,A) < FAC
A = 0 if (Y,A) = FAC
A = FF if (Y,A) > FAC

NEGOP EED0
FAC = -FAC

FADDH E7A0
Add 1/2 to FAC

DIV10 EA55
Divide FAC by 10. Returns positive numbers only.

MUL10 EA39

Multiply FAC by 10. Works for both positive and negative numbers.

Apple Tech Notes

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